MINI PROJECT REPORT

on

KAVACH AI

Voice-Enabled Scam & Fraud Message Detector with Explainable ML

Submitted in partial fulfillment of the requirements of the degree of

Bachelor of Engineering in

Information Technology by

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Department of Information Technology
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2026-27

CERTIFICATE

This is to certify that the Data Science Lab Mini project entitled **"KAVACH AI"** is a bonafide work of the following students, submitted to the University of Mumbai in partial fulfillment of the requirement for the award of the degree of **Bachelor of Engineering** in **Information Technology.**

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PROJECT REPORT APPROVAL

This Data Science Lab Mini project report entitled ***KAVACH AI*** by following students is approved for the degree of ***Bachelor of Engineering*** in ***Information Technology.***

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Place:

DECLARATION

I declare that this written submission represents my ideas in my own words and where others ideas or words have been included, I have adequately cited and referenced the original sources. I also declare that I have adhered to all principles of academic honesty and integrity and have not misrepresented or fabricated or falsified any idea/data/fact/source in my submission. I understand that any violation of the above will be cause for disciplinary action by the Institute and can also evoke penal action from the sources which have thus not been properly cited or from whom proper permission has not been taken when needed.

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Date: 21-10-2026

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ABSTRACT

Fraudulent SMS and WhatsApp messages are a daily problem in India: fake bank KYC alerts, "digital arrest" calls from people posing as the police or the CBI, part-time job and investment offers that ask for a fee, and parcel or customs scams. They are written to create panic and urgency, and many people cannot easily tell them apart from genuine bank or delivery notifications.

Kavach AI is a voice-enabled scam and fraud message detector with explainable machine learning. It is trained on the Sentinel multiclass dataset (14,000 messages) and classifies a message into five classes: legit, government impersonation, financial fraud, bank phishing, and romance/parcel/sextortion. TF-IDF features (word 1-2 grams and character 3-5 grams) are compared across Naive Bayes, Logistic Regression, a calibrated Linear SVM and a Random Forest. Because the dataset is heavily templated (only 6,693 distinct templates), every model is evaluated on a template-grouped split as well as the usual random split. The Linear SVM is the best model, with a macro-F1 of 0.9854 on the grouped split; the random split's 0.9976 is inflated by repeated templates. On out-of-domain data the model never saw, the scam-vs-legit F1 is 0.288 on the UCI SMS Spam Collection, 0.361 on a Hinglish & English scam set and 0.756 on its 87 Hinglish-only messages; the report explains the reasons for this gap.

A Streamlit web application has four pages: Home, Scan a Message (pasted text, a screenshot read by RapidOCR, or a voice note transcribed by faster-whisper), Model Performance and Kavach Assistant. Every verdict shows the scam probability, a High/Medium/Low risk tier, the words that drove the prediction (LIME), a red-flag checklist from a separate English and Hinglish rule layer, and vetted next steps (helpline 1930 and cybercrime.gov.in). The assistant answers by text or voice in English, Hindi or Hinglish using a local language model (qwen2.5:7b through Ollama), but the trained model, not the language model, makes every classification. The whole system runs locally on a laptop, so no message ever leaves the device.

INTRODUCTION

Digital payments and smartphone messaging are now part of everyday life in India, and message-based fraud has grown with them. The Government of India runs the national cyber-crime helpline 1930 and the National Cyber Crime Reporting Portal (cybercrime.gov.in), and public advisories repeatedly warn that the police or the CBI never arrest anyone over a video call. Even so, the first line of defence is the person reading the message, often at a moment when the message is designed to make them panic.

Fraud messages share recognisable tactics: urgency ("today", "within 30 minutes"), threats (an account blocked, a connection cut, an arrest warrant), requests for an OTP, PIN or fee, links to look-alike websites, and claims of authority. Spotting these tactics takes attention and awareness that many users lack, especially elderly and first-time smartphone users. The messages also arrive in English, Hindi and Hinglish (Hindi written in Roman script), and often as a screenshot or a forwarded voice note rather than plain text.

This mini project builds Kavach AI ("kavach" means armour or shield in Hindi), which combines three ingredients. First, a supervised machine-learning classifier that assigns every message to one of five classes with a calibrated scam probability. Second, explainability: LIME highlights the words that drove each prediction, and an independent rule layer lists the scam tactics it finds. Third, accessible input and output: the user can paste text, upload a screenshot or speak, and a voice-enabled assistant explains the result in the user's

own language.

Two design principles run through the project. The trained model is the classifier and the language model only explains its result, which keeps every verdict measurable and reproducible. And the model is evaluated honestly: besides the usual random split, it is tested on a template-grouped split and on two out-of-domain datasets, and the limitations found are reported rather than hidden. The entire system runs locally on a laptop, so personal messages stay on the device and there is no cloud or API cost.

LITERATURE SURVEY

SMS spam and fraud detection has been studied as a text-classification problem for over a decade. Early work focused on English spam filtering with bag-of-words features and linear classifiers; more recent work adds explainability, careful evaluation and, with large language models, conversational interfaces. The studies below shaped the design of Kavach AI.

T. A. Almeida, J. M. Gómez Hidalgo and A. Yamakami (ACM DocEng, 2011) released the SMS Spam Collection of 5,574 English messages and compared several classifiers on it, with linear SVMs among the strongest. The collection is used in this project as an out-of-domain test set [1].

S. J. Delany, M. Buckley and D. Greene (Expert Systems with Applications, 2012) reviewed SMS spam filtering methods and datasets and observed that content-based classifiers work well but the public datasets are small and region-specific. This motivates testing a model on data from a different source, as done here [2].

T. Joachims (ECML, 1998) showed that support vector machines suit text categorisation because text features are high-dimensional and sparse, and that linear SVMs outperformed Naive Bayes, k-nearest neighbours and decision trees. This is the basis for the TF-IDF and linear-SVM pipeline [3]. J. Platt (1999) proposed fitting a sigmoid to SVM scores to obtain calibrated probabilities, which Kavach uses for its scam probability and risk tiers [4].

M. T. Ribeiro, S. Singh and C. Guestrin (ACM KDD, 2016) introduced LIME, which explains a single prediction of any classifier by perturbing the input and fitting a simple local model. Kavach uses LIME to highlight the words behind each verdict [5].

S. Kaufman, S. Rosset, C. Perlich and O. Stitelman (ACM TKDD, 2012) formalised data leakage, where information that would not be available at prediction time leaks into training and inflates evaluation scores. Near-identical templated messages in both the training and test sets are a form of leakage, which motivated the template-grouped split used here [6].

A. Radford et al. (ICML, 2023) presented Whisper, a speech-recognition model trained on 680,000 hours of weakly supervised multilingual audio that works robustly without fine-tuning. Kavach's voice input uses faster-whisper, an efficient reimplementation of Whisper [7], [14].

S. Yao et al. (ICLR, 2023) showed that language models can interleave reasoning with tool calls (ReAct), and K. Greshake et al. (ACM AISec, 2023) showed that applications which let a language model process untrusted text are vulnerable to indirect prompt injection. Kavach's assistant therefore calls the classifier from code rather than leaving it to the language model, treats every message as data, and checks every reply with code-level guards [8], [9].

PROBLEM DEFINITION

The primary dataset is the Sentinel multiclass scam dataset [10]: 14,000 messages with a text column and an integer label mapped to five classes: legit (5,600), government impersonation including "digital arrest" (2,905), financial fraud (3,080), bank phishing (1,875) and romance/parcel/sextortion (540). The classes are imbalanced: the smallest class is only 3.9% of the data, against 40% for legit.

The task is: given a suspicious message as text, a screenshot or a voice note, predict its class; give a scam probability, defined as 1 - P(legit), with a risk tier (High at 0.80 or above, Medium from 0.50 to 0.80, Low below 0.50); explain which words and tactics drove the verdict; and tell the user what to do next. Because of the class imbalance, models are compared on macro-F1, which weights every class equally, rather than on accuracy alone.

Three properties of the data make honest evaluation part of the problem. The dataset has no exact duplicates but only 6,693 distinct templates once digits, links, phone numbers and amounts are masked, so a random split puts near-identical messages in both training and test sets. Its legit class consists of service notifications (banks, recharges, deliveries) with no casual personal chat. And it covers only four scam types. The model is therefore also tested out-of-domain on the UCI SMS Spam Collection [1] (5,171 messages after removing 403 duplicates) and a Hinglish & English scam dataset [11] (3,558 after removing 229 duplicates). As 97.6% of the latter is the UCI collection, its 87 rows that are not in UCI (47 scams) are reported separately as the Hinglish-only test.

There is also an operational problem. The users are non-technical, many prefer to forward a screenshot or speak rather than type, and their messages may contain personal and banking details. The system must therefore accept text, images and voice in English, Hindi and Hinglish, run entirely on the user's own machine, and never let the conversational assistant invent a helpline number or contradict the model.

PROPOSED METHODOLOGY

Data preparation. The three datasets are downloaded by scripts/download_data.py. Sentinel's integer labels are mapped to class names using its dataset card. Exact duplicates are checked in every dataset (none in Sentinel; 403 removed from UCI and 229 from the Hinglish set). Text is lowercased, links are replaced by a <URL> token, money amounts (₹, Rs or INR followed by digits) by an <AMOUNT> token, and whitespace is collapsed. Hinglish words are kept as they are, with no stemming or stop-word removal.

Engineered features. For the analysis and the red-flag checklist, each message also gets has_url, has_phone, has_amount, an urgency-word count and its length. These features are used for exploratory analysis and display; the classifier itself learns from the text.

Phone-number shortcut. Phone numbers occur in 16.2% of Sentinel's legit messages and 0.0% of its scam messages, an accident of how the dataset was built, since real scams usually include a number. A <PHONE> token would teach the model that a phone number means legit, so phone numbers are removed from the model's input. They are still detected and shown in the checklist.

Evaluation splits. Random split: stratified 80/20 with random_state 42 (11,200 training and 2,800 test messages). Template-grouped split: each message gets a template key (the cleaned text with digits, links,

phone numbers and amounts masked), and StratifiedGroupKFold (5 folds, one fold held out) keeps every template entirely on one side of the split (11,200 / 2,800 messages, 6,693 templates). The best model is chosen on the grouped split.

Features. A union of two TF-IDF vectorisers: word 1-2 grams capture phrases such as "KYC expired" or "arrest warrant", and character 3-5 grams (within word boundaries) handle the spelling variation common in Hinglish and in deliberately misspelt scams. Both use min_df 2 and sublinear term frequency.

Models. Four classifiers are compared: Multinomial Naive Bayes (alpha 0.1); Logistic Regression (balanced class weights, C 10); a Linear SVM (balanced class weights, C 1) wrapped in CalibratedClassifierCV with sigmoid calibration and 3 folds, so that it outputs probabilities [4]; and a Random Forest with 200 trees and balanced class weights. The final Linear SVM is refit on all 14,000 messages and saved as artifacts/ model.joblib (2.5 MB).

Explainability. Local explanations come from LIME [5] with 500 perturbed samples per message (about 0.03 s), returned as (word, weight) pairs and highlighted in the message: red words push toward the scam class, green words toward legit. Global explanations are the top word n-grams per class from the Linear SVM coefficients, averaged over the three calibration folds.

Rule layer. A separate set of hand-written patterns in English and Hindi/Hinglish detects nine scam tactics: threats to disconnect or block, urgency, payment requests, OTP/PIN/password requests, "call this number", links, secrecy, "new number" impersonation, and claims of authority. Three or more tactics give a "high" advisory and two give "caution". The rule layer is an independent second signal: it never changes the model's prediction, probability or risk tier. The verdict card combines the two signals as shown in Table 1.

Table 1 - How the verdict card combines the model and the rule layer

Model prediction	Rule-layer advisory	Card shown to the user
Any scam class	any	Red/orange: "Likely scam: <class>" with the risk tier
Legit	high (3+ flags)	Amber: "Be careful: N warning signs found"
Legit	caution (2 flags)	Light amber: "No scam pattern matched, but check: ..."
Legit	none	Green: "No scam pattern detected" (never "safe")

Under every card the raw model output is shown in small text (for example "Model: legit · scam probability 10%"), so the model's own verdict is never hidden.

SYSTEM ARCHITECTURE:

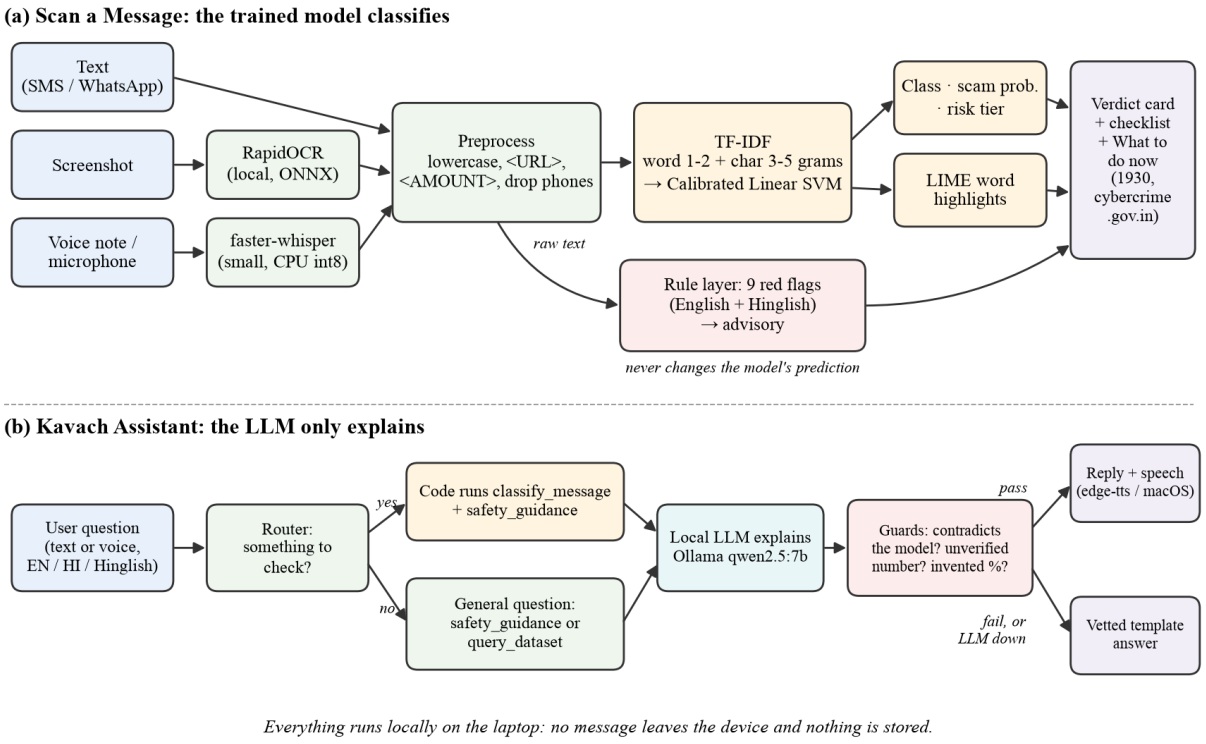


Fig. 1 - System architecture: (a) the scan pipeline, where the trained model classifies; (b) the Kavach Assistant, where the local language model only explains.

Serving layer. The web application is a multipage Streamlit app. Screenshots are read by RapidOCR (an ONNX OCR engine) [15] and voice notes are transcribed by faster-whisper using the small model on the CPU with int8 weights [14]; in both cases the extracted text is shown in an editable box before it is classified. The trained model is loaded once and cached; OCR and speech models load only when their tab is first used.

Assistant. The Kavach Assistant uses qwen2.5:7b [12] served locally by Ollama [13] through an OpenAI-compatible API, with three tools: classify_message (the trained model plus LIME and the rule layer), query_dataset (counts, percentages and examples from Sentinel) and safety_guidance (static, vetted advice). A router in code decides whether the input contains something to check (a message, link, phone number, amount or described incident); if so, the code itself calls classify_message and safety_guidance before the language model is involved, so the verdict never depends on the language model choosing a tool. General questions such as "What is a digital arrest scam?" are never classified and get no probability.

Guards and fallback. Every reply is checked in code. It is replaced by a vetted template built from the tool outputs if it calls a flagged message safe, contains a phone number or website that did not come from a tool or the user, quotes a probability no tool produced, repeats itself, or uses the wrong script. If Ollama is not running or times out, the same template answer is shown. Replies are spoken with edge-tts (en-IN-NeerjaNeural, or hi-IN-SwaraNeural for Hindi) and, without internet, with the built-in macOS voices (Rishi and Lekha).

CODE AND RESULT

The code is organised as a Python package. src/preprocess.py cleans text and builds the template keys; src/train.py trains the four models on both splits; src/evaluate.py computes in-domain and out-of-domain metrics, the EDA statistics and the rule-layer measurements and writes artifacts/metrics.json and the figures; src/explain.py produces the LIME and global explanations; src/predict.py, src/red_flags.py and src/guidance.py are used by the app. Every number in the app and in this report is read from artifacts/metrics.json. Each model trains in under 3 seconds on the project laptop (Apple M5, 16 GB).

Exploratory data analysis

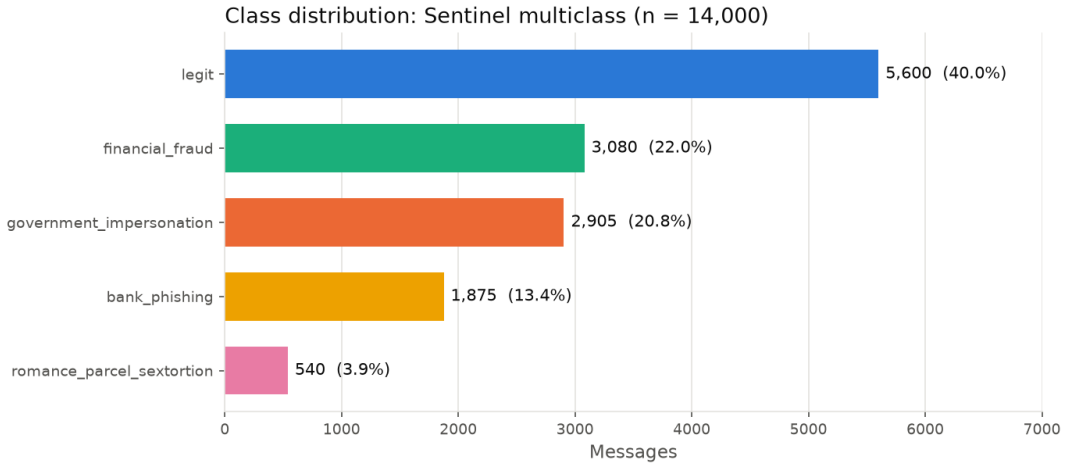


Fig. 2 - Class distribution of the Sentinel training data.

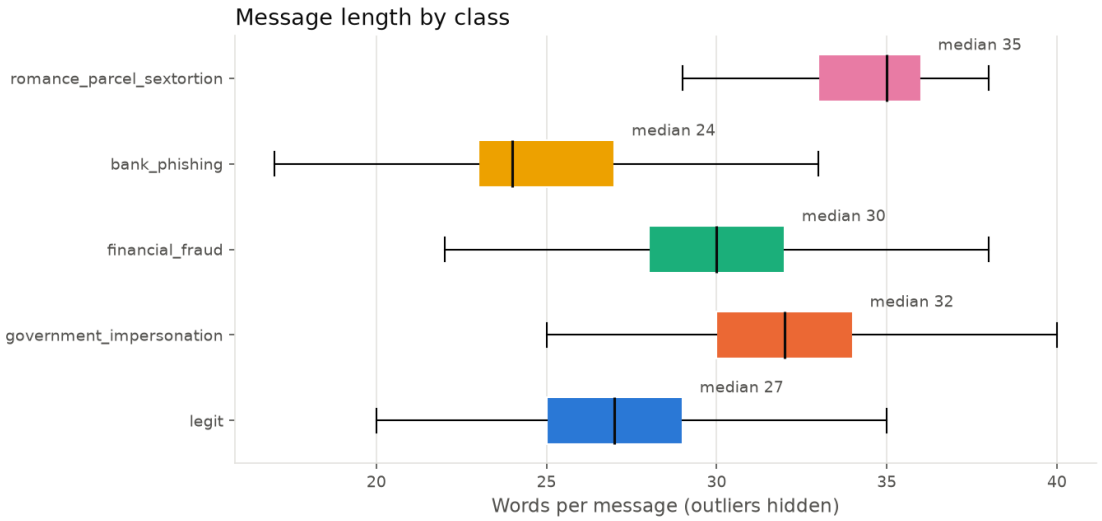


Fig. 3 - Message length by class.

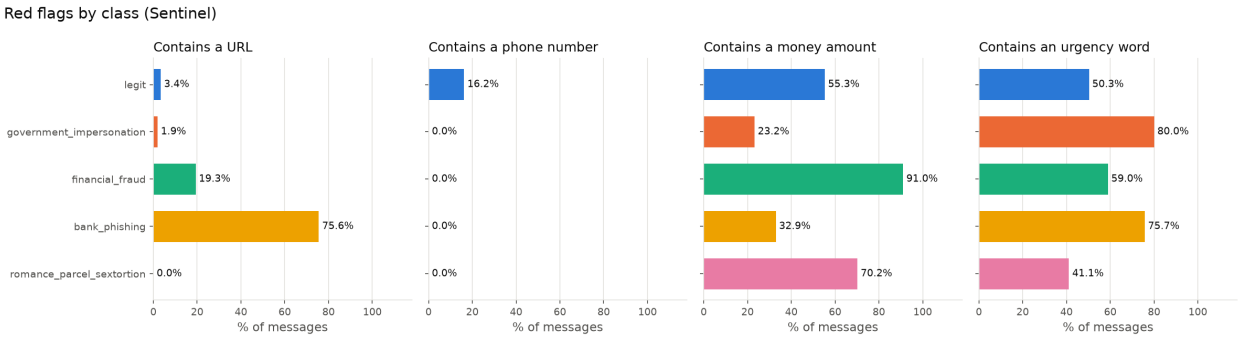


Fig. 4 - Share of messages per class containing a URL, a phone number, a money amount or an urgency word.

Links appear in 24.6% of scam messages against 3.4% of legit ones, and in 75.6% of bank-phishing messages. Money amounts are frequent everywhere (53.3% of scam and 55.3% of legit messages, because legit messages are transaction alerts), but reach 91.0% in financial fraud. Urgency words appear in 80.0% of government-impersonation messages. Scam messages are slightly longer on average (29.8 words against 26.9). The phone-number column shows the dataset artefact described in the methodology.

Model comparison

Table 2 - Model leaderboard (macro-F1 on the 2,800-message test sets)

Model	Macro-F1 (grouped)	Macro-F1 (random)	Accuracy (grouped)	ROC-AUC OvR (grouped)	Train time (s)
Linear SVM (best)	0.9854	0.9976	0.9854	0.9995	2.8
Logistic Regression	0.9573	0.9968	0.9679	0.9993	1.8
Random Forest	0.9538	0.9824	0.9611	0.9958	1.5
Naive Bayes	0.9388	0.9756	0.9486	0.9947	1.0

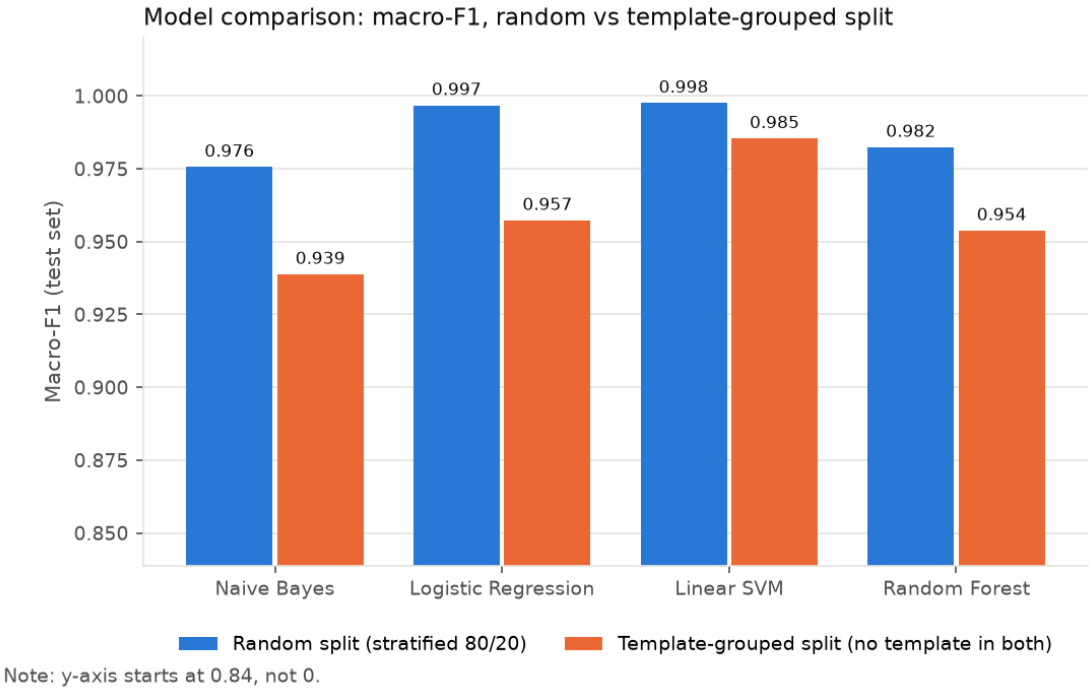


Fig. 5 - Macro-F1 of the four models on the random and template-grouped splits (the y-axis starts at 0.84).

The Linear SVM has the highest macro-F1 on both splits and is the selected model: 0.9854 on the grouped split, ahead of Logistic Regression (0.9573), Random Forest (0.9538) and Naive Bayes (0.9388). Every model scores lower on the grouped split than on the random split, which confirms that the random split rewards memorised templates. Even the grouped score is an upper bound, because templates that differ only in a bank or agency name still count as different templates.

Table 3 - Per-class results of the Linear SVM

Class	Precision (grouped)	Recall (grouped)	F1 (grouped)	F1 (random)	Test support
legit	0.9991	1.0000	0.9996	0.9996	1120
government_impersonation	0.9626	0.9759	0.9692	0.9983	581
financial_fraud	0.9818	0.9610	0.9713	0.9968	616
bank_phishing	0.9815	0.9920	0.9867	0.9933	375
romance_parcel_extortion	1.0000	1.0000	1.0000	1.0000	108

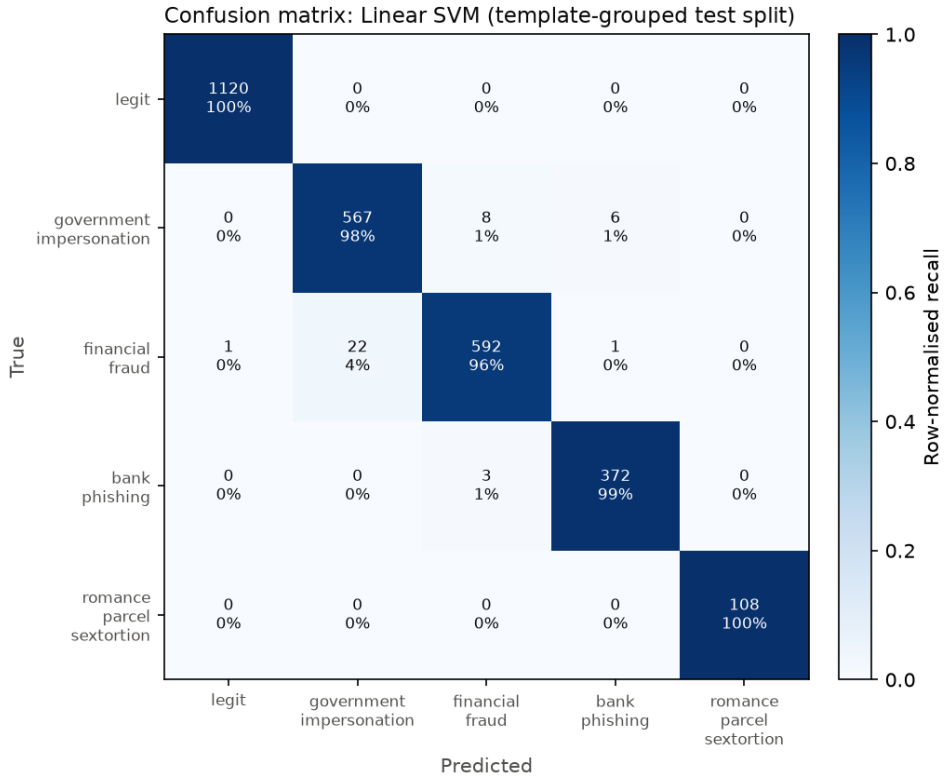


Fig. 6 - Confusion matrix of the Linear SVM on the template-grouped test split.

Almost every test message lies on the diagonal. All 1,120 legit test messages are classified correctly, and the smallest class, romance/parcel/sextortion, reaches an F1 of 1.0 thanks to balanced class weights. The main confusion is between financial fraud and government impersonation (22 messages), which matches the label noise found in the data: "PNB fraud alert" templates appear under three classes, NCB impersonation messages are labelled financial fraud, and almost all parcel and customs scams are labelled financial fraud rather than romance/parcel.

Out-of-domain evaluation

Table 4 - Out-of-domain results (scam vs legit; positive class = scam)

Test set	n	Accuracy	Precision	Recall	F1
UCI SMS Spam	5,171	0.7142	0.2101	0.4579	0.2881
Hinglish & English (full)	3,558	0.6928	0.2918	0.4739	0.3612
Hinglish-only (not in UCI)	87	0.7701	0.8857	0.6596	0.7561

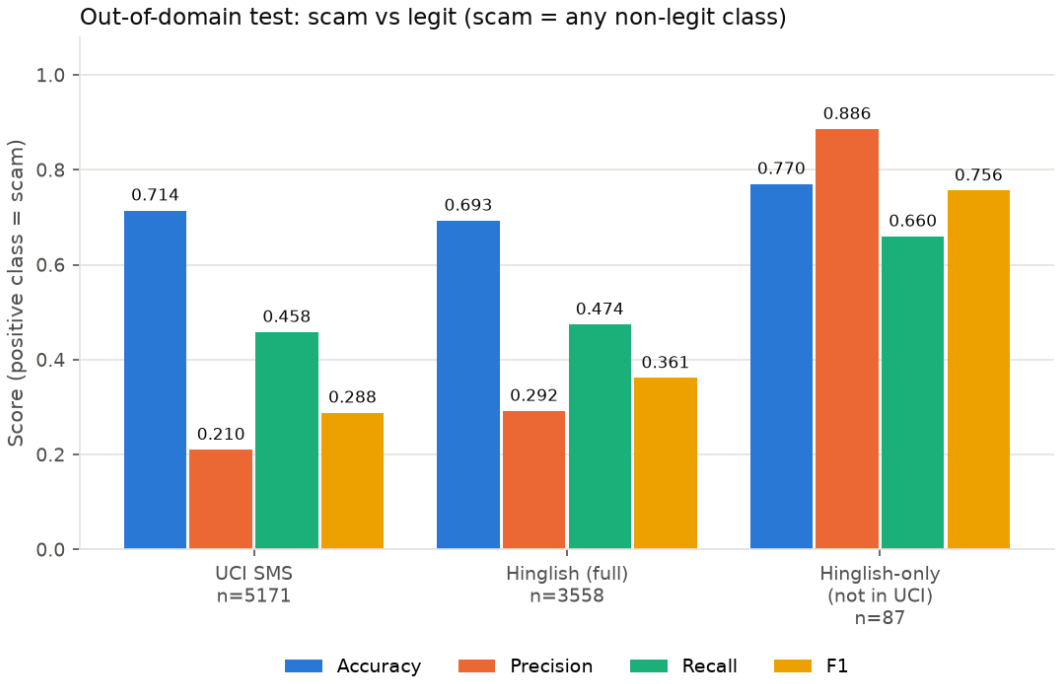


Fig. 7 - Out-of-domain test results.

These scores are much lower than the in-domain ones, for three reasons found in the analysis. First, Sentinel's legit class has no casual chat, so ordinary personal messages are often flagged as scams, which explains the low precision on UCI (0.21). Second, UCI's spam is UK prize and lottery spam that looks little like Indian fraud, so recall is 0.46. Third, the model partly learned shortcuts such as bank and agency names (Fig. 8). On real Hinglish scams it does better (F1 0.76), but with only 87 messages that estimate has a wide margin of error. No tuning was done on these sets, so they remain a fair test.

Phone-number ablation and rule layer

Table 5 - Effect of removing phone numbers from the model's input (Linear SVM)

Model input	Macro-F1 (grouped)	UCI F1	UCI recall	Hinglish-only F1
With a <PHONE> token	0.9851	0.2422	0.3783	0.7561
Phone numbers removed (final)	0.9854	0.2881	0.4579	0.7561

Removing phone numbers leaves the in-domain score unchanged but raises real-world recall on UCI from 0.38 to 0.46, confirming that the token was a shortcut rather than a useful signal.

Table 6 - Rule layer: false alarms on legit messages and scam recall

Measurement	n	Value
Sentinel legit (test split): any advisory	1120	6.16%
Sentinel legit (test split): "high" advisory	1120	0.00%
UCI legit: any advisory	4,518	0.04%
UCI legit: "high" advisory	4,518	0.00%
Hinglish-only scam recall: model only	47	0.6596
Hinglish-only scam recall: model or "high" advisory	47	0.6809

The rule layer is cautious: no legit message in either dataset receives a "high" advisory, and only 6.2% of Sentinel's legit messages get any warning. It adds little recall at the "high" threshold on the out-of-domain sets, but it catches scam types the model has never seen, such as the Hinglish electricity-disconnection scam in Fig. 10. The rules were written from general scam-awareness guidance without looking at the out-of-domain rows.

Explainability

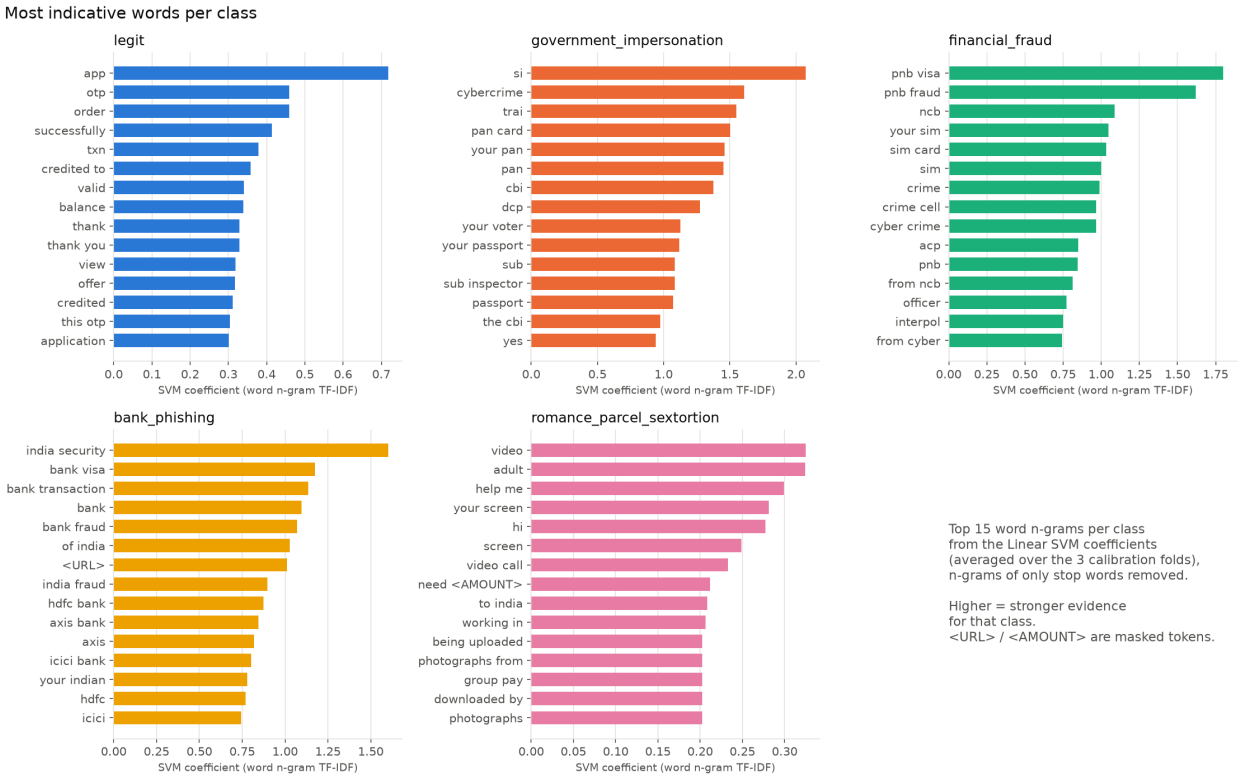


Fig. 8 - Most indicative word n-grams per class (Linear SVM coefficients).

The government-impersonation words (cybercrime, TRAI, PAN card, CBI, sub inspector) and the romance words (video, adult, help me, your screen) describe real scam tactics. The financial-fraud and bank-phishing lists, however, are led by bank and agency names ("pnb visa", "india security", "hdfc bank"), which shows that the model partly learned which names each template uses. This is a key reason for the out-of-domain gap

and is listed as future work.

Demo results and application screenshots

On the 14 fictional demo messages, the model predicts the exact class for 11 and scam-vs-legit for 12. The parcel scam is predicted as financial fraud (still High risk), matching the label noise above, and the two Hinglish scams are predicted legit but receive the amber "Be careful: 4 warning signs found" card from the rule layer. On the project laptop, OCR takes about 0.3-0.4 s per screenshot, Whisper about 2.5 s per voice clip, LIME about 0.03 s per message, and an assistant reply typically 10-30 s. The screenshots below are taken directly from the running application.

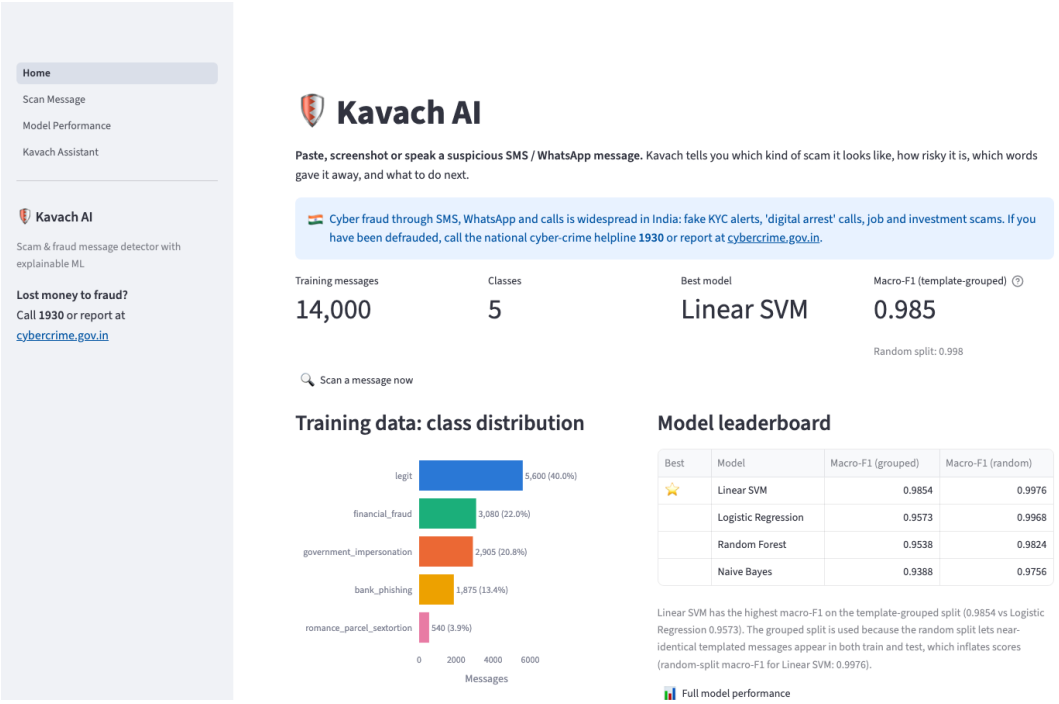


Fig. 9 - Home page: dataset KPIs, the grouped and random macro-F1, class distribution and the model leaderboard.

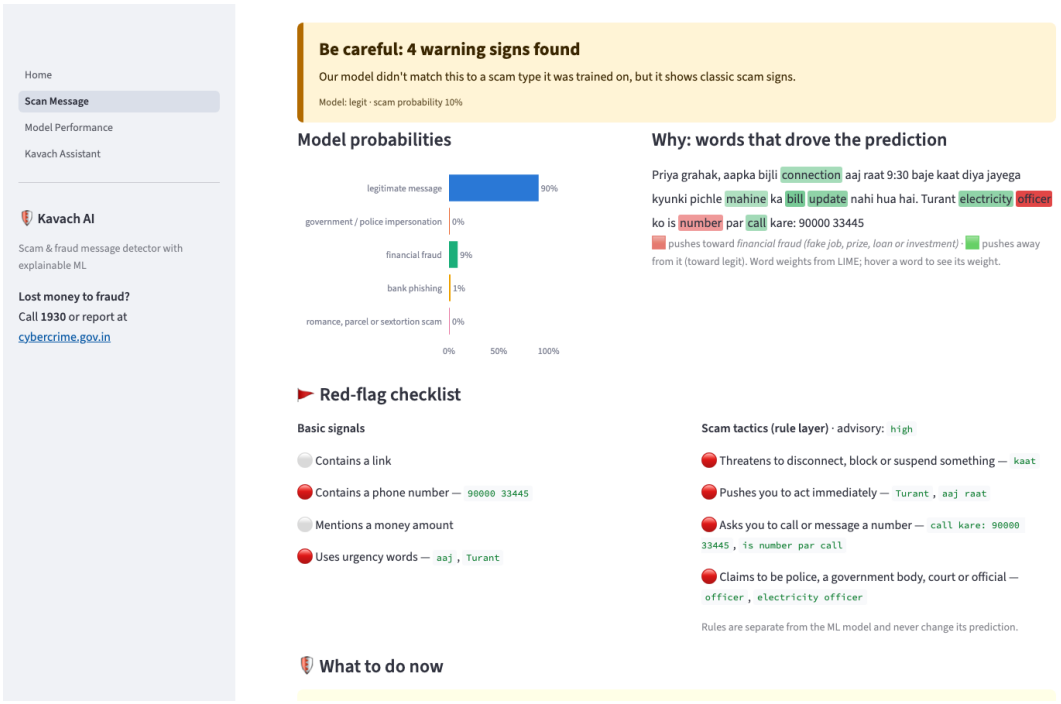


Fig. 10 - Scan a Message (text): a Hinglish electricity-disconnection scam. The model says legit (scam probability 10%), but the rule layer finds 4 warning signs and the card turns amber.

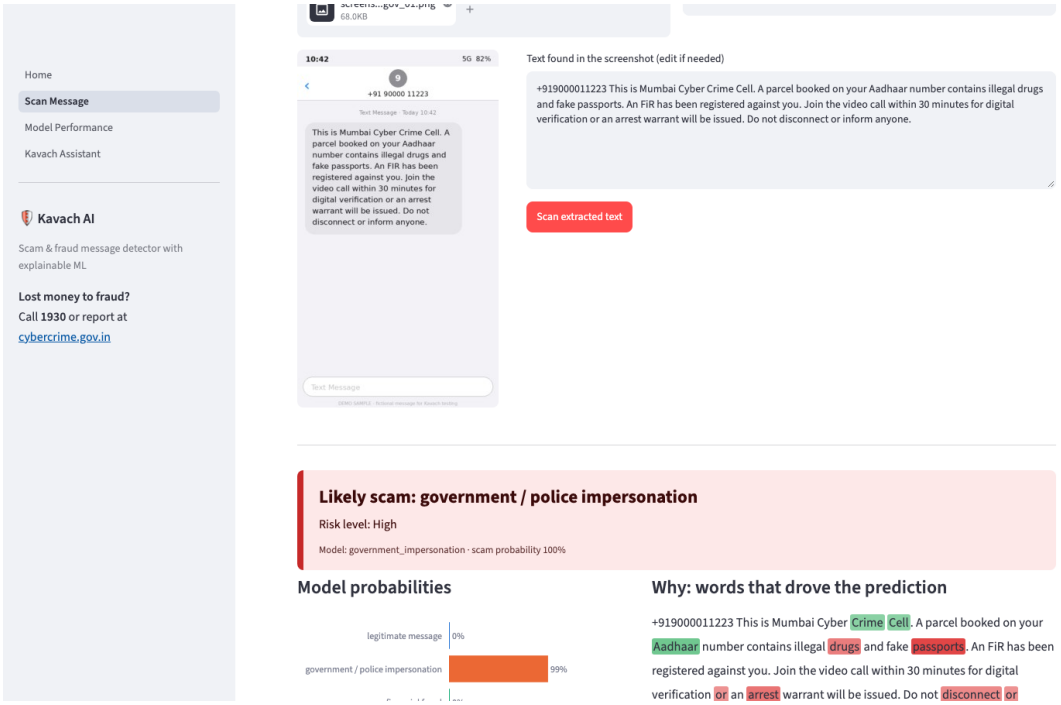


Fig. 11 - Scan a Message (screenshot): RapidOCR reads a "digital arrest" message, classified as government / police impersonation with High risk; LIME highlights the trigger words.

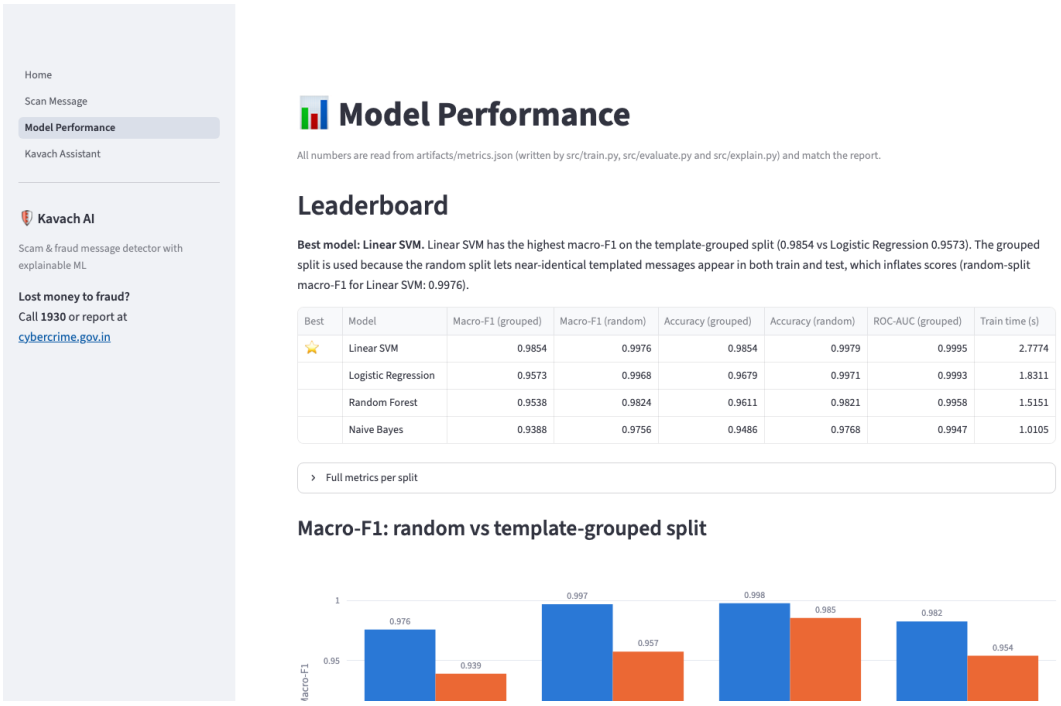


Fig. 12 - Model Performance page: leaderboard for both splits and the macro-F1 comparison.

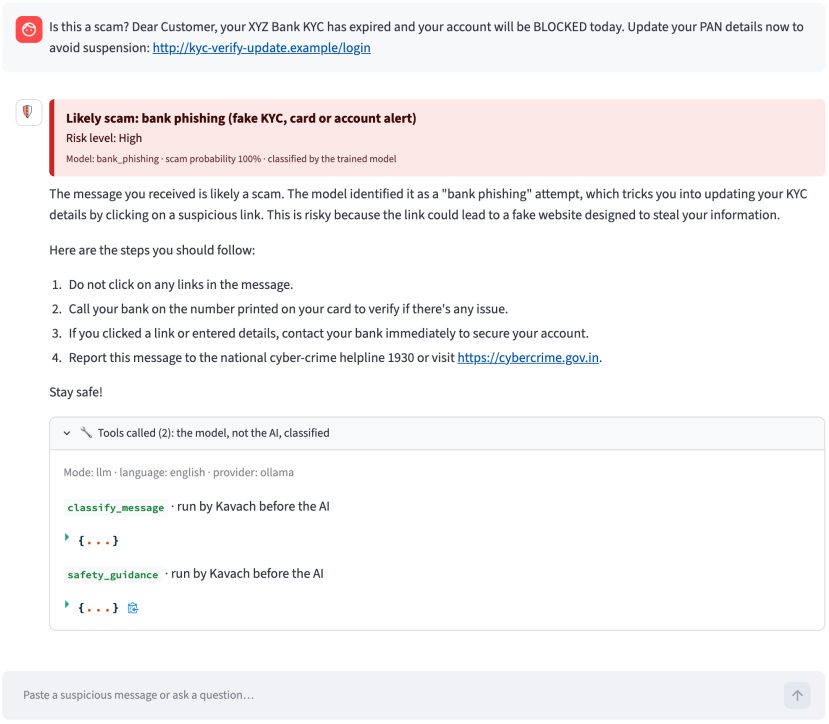


Fig. 13 - Kavach Assistant answering a bank-phishing message in English. The tools panel shows that classify_message and safety_guidance were run by Kavach before the language model.

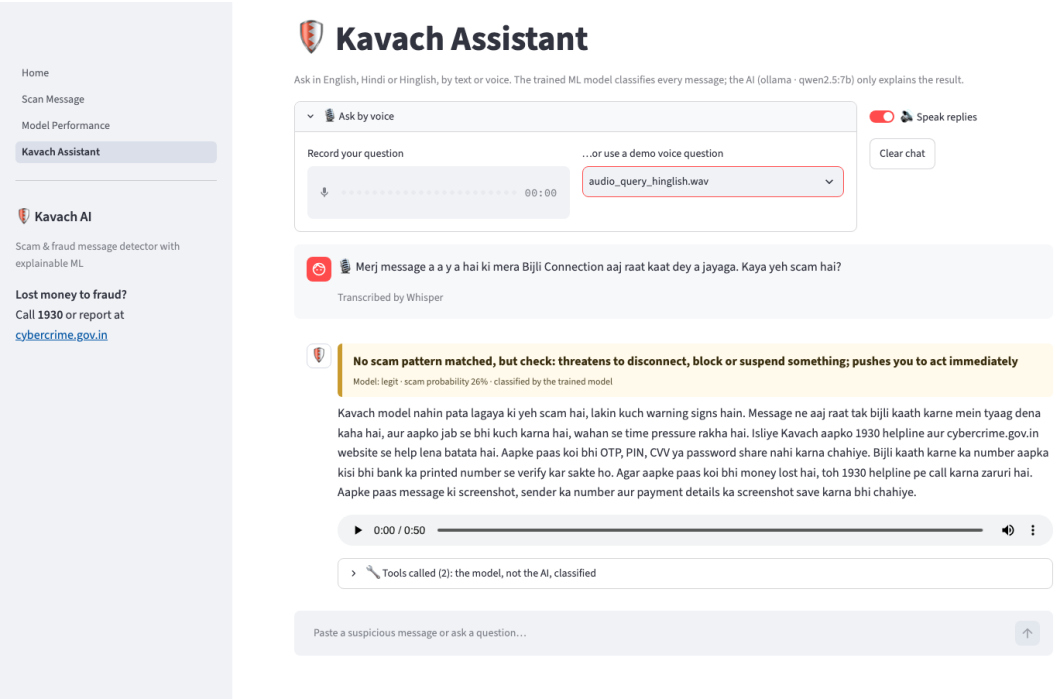


Fig. 14 - Kavach Assistant answering a Hinglish voice question transcribed by Whisper, with a spoken reply.

BUSINESS INTELLIGENCE DECISION

Kavach turns the model output into a concrete decision for the user through three risk tiers derived from the scam probability, combined with the rule-layer advisory.

High (scam probability 0.80 or above). Treat the message as a scam: do not click, call, reply or pay; block the sender; report it on 1930 or cybercrime.gov.in; and, if money has already been lost, call 1930 immediately and inform the bank.

Medium (0.50 to 0.80). Do not act on the message until it has been verified through an independent, official channel, such as the number printed on the user's bank card.

Low (below 0.50). No scam pattern detected. This is never presented as "safe": if the rule layer finds warning signs, the card turns amber and the user is told to treat the message as a possible scam.

The advice is class-specific and comes from fixed, vetted text rather than from the language model:

Government impersonation: police, the CBI or any other agency never arrest or question anyone over a video call; disconnect, do not pay any "verification" amount, and tell a family member.

Bank phishing: do not open KYC links from an SMS; banks do not ask for an OTP, PIN or password; call the number printed on the card.

Financial fraud: do not pay a registration, processing or "tax" fee to receive a job, prize, loan or refund; guaranteed high returns are a classic sign of investment fraud.

Romance, parcel or sextortion: do not pay customs or delivery charges for an unexpected gift, do not send money to someone met only online, and never pay a sextortion threat.

The data analysis also supports awareness decisions. Links are seven times more common in scams (24.6%

against 3.4%) and appear in three quarters of bank-phishing messages, so "never open a link from an SMS" is the single most useful message for awareness campaigns. Financial fraud almost always names an amount (91.0%), and government impersonation relies on urgency (80.0%). Through the query_dataset tool the assistant answers such questions with exact figures; for example, "How many bank phishing messages are in the dataset?" returns 1,875 (13.39%).

APPLICATION

Kavach AI can be used directly in several settings.

Personal scam checker. Citizens, especially elderly and first-time smartphone users, can paste, screenshot or simply speak a suspicious message and get a verdict, an explanation and safe next steps in seconds.

Bank and telecom customer support. A support agent can check a message forwarded by a worried customer and explain, with the highlighted words and red flags, why it is or is not a known scam pattern.

Cyber-cell awareness and training. The explanations, red-flag checklists and per-class statistics make practical material for awareness drives in colleges, housing societies and offices.

Digital-literacy sessions. Because every verdict shows why it was reached, the tool teaches people to recognise the tactics themselves rather than only trusting a label.

Privacy-sensitive deployments. Because everything runs locally, the same system can be used where messages must not leave the device or the organisation.

The pipeline and its honest-evaluation method (grouped split, out-of-domain tests, shortcut ablation) also carry over to other short-text fraud problems, such as phishing e-mails or fraudulent app reviews.

CONCLUSION

This project built Kavach AI, a voice-enabled scam and fraud message detector with explainable machine learning. It combines a TF-IDF and calibrated Linear SVM classifier, LIME explanations, an independent English and Hinglish rule layer, and a Streamlit application that accepts text, screenshots and voice, with a local voice assistant that explains each result.

The Linear SVM was the best of four models, with a macro-F1 of 0.9854 on a template-grouped split. The project shows why that number, and not the random split's 0.9976, should be reported, and why out-of-domain tests are needed: on data from other sources the F1 falls to 0.29 (UCI) and 0.76 (Hinglish-only). The analysis traced this gap to templated training data, bank-name shortcuts, the absence of casual chat in the legit class and label noise, and removed one shortcut (phone numbers), which raised real-world recall.

The assistant keeps the trained model as the only classifier: the code runs the model before the language model answers, and code-level guards stop any reply that contradicts the model, invents a helpline or quotes an unsupported probability. The whole system runs on a laptop, so users' messages never leave the device.

Future work includes fine-tuning a multilingual transformer (distilbert-base-multilingual-cased) and comparing it with the Linear SVM; collecting a real Hinglish dataset that includes casual legit chat; adding scam types that are missing today, such as utility-disconnection and "new number" scams; relabelling the noisy classes; adding a user-feedback loop for retraining; and building a mobile or message-forwarding

version so that people can check messages directly from their phones.

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